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Review article

Using multimedia for e-learning

R.E. Mayer 

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Abstract

This paper reviews 12 research-based principles for how to design computer-based multimedia instructional materials to promote academic learning, starting with the multimedia principle (yielding a median effect size of $d = 1.67$ based on five experimental comparisons), which holds that people learn better from computer-based instruction containing words and graphics rather than words alone. Principles aimed at reducing extraneous processing (i.e., cognitive processing that is unrelated to the instructional objective) include coherence ($d = 0.70$), signalling ($d = 0.46$), redundancy ($d = 0.87$), spatial contiguity ($d = 0.79$) and temporal contiguity ($d = 1.30$). Principles for managing essential processing (i.e., mentally representing the essential material) include segmenting ($d = 0.70$), chunking ($d = 0.70$) and chunking ($d = 0.70$). Principles for fostering generative processing (i.e., making sense of the material) include personalization ($d = 0.70$), voice ($d = 0.70$) and

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Lay Description

Coherence principle: Eliminate extraneous material.

Signaling principle: Highlight essential material.

Redundancy principle: Do not add on-screen text to narrated graphics.

Spatial contiguity principle: Please printed words next to corresponding graphics.

Temporal contiguity: Present narration simultaneously with corresponding graphics.

Segmenting principle: Break lesson into to self-paced parts.

Pre-training principle: Provide pre-training in key terms.

Modality principle: Use spoken text rather than printed text with graphics.

Personalization principle: Use conversational language.

Voice principle: Use appealing human voice.

Embodiment: Show on-screen agents that use human-like gestures.

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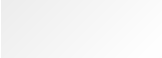
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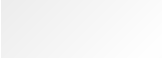
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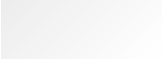
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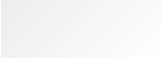
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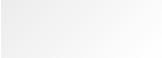
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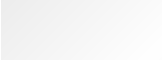
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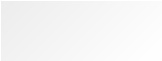
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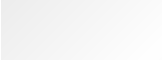
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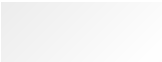
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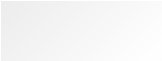
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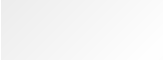
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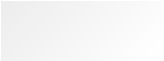
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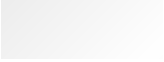
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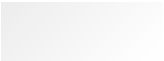
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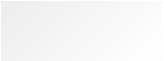
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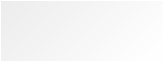
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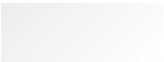
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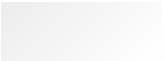
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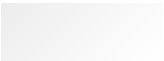
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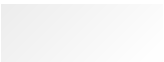
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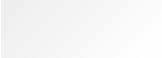
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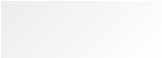
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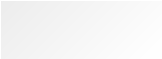
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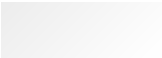
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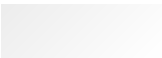
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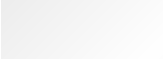
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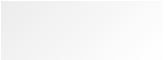
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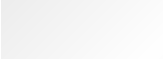
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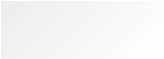
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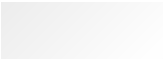
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